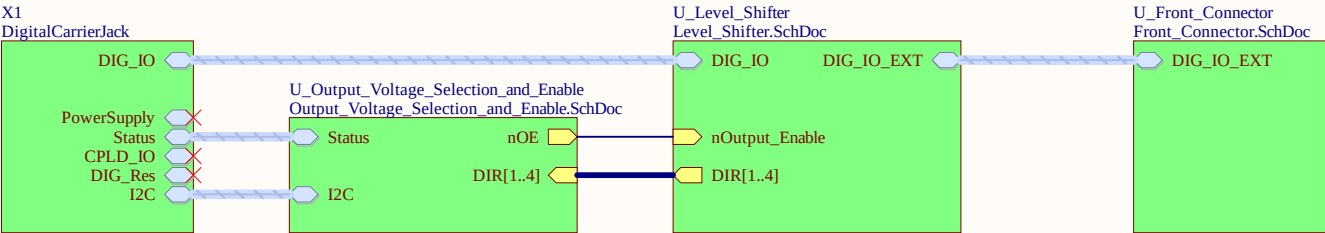
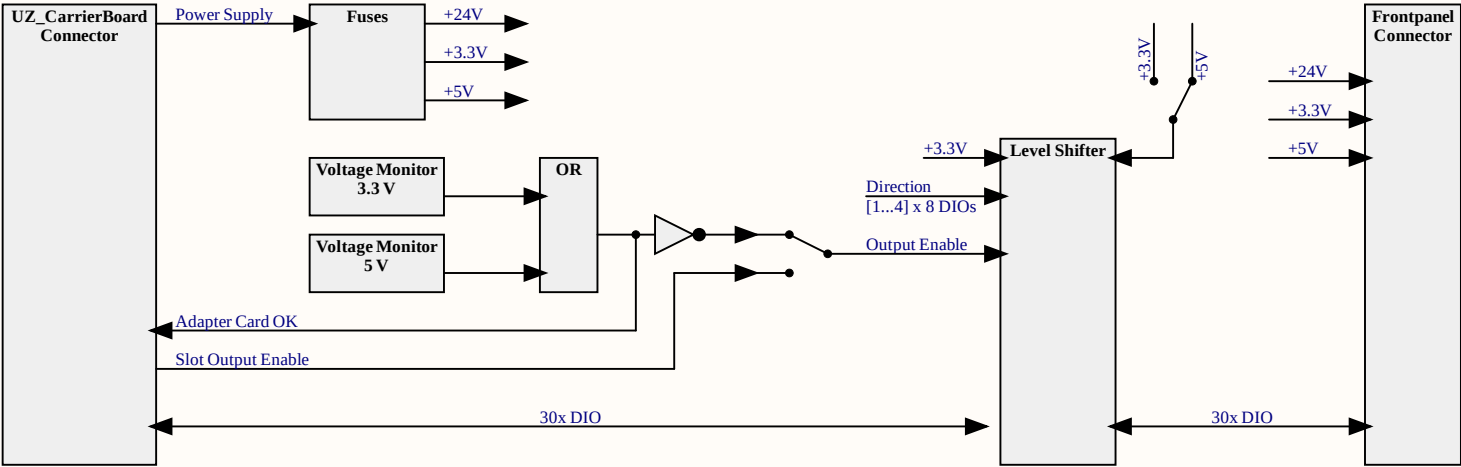


Top Sheet



Block Diagram

- Board Features:
- configurable output voltage level (3.3V or 5V)
 - voltage monitor ICs supervise 3.3V and 5V supply rail
 - output enable of level shifters can be controlled either by the supervisors status or via UZ_CarrierBoard signal
 - I²C-Port Expander reads selected output voltage level and signal direction per group of 8 channels
 - RGB-LED at front panel visualizes selected voltage and error status



Chose size of UZ logo
by chosing footprint
type



LOG01

INFO

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

Title TopSheet.SchDoc

Revision: Rev02

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

UltraZohm

www.ultrazohm.com

Date: 03.04.2025

Sheet 1 of 5

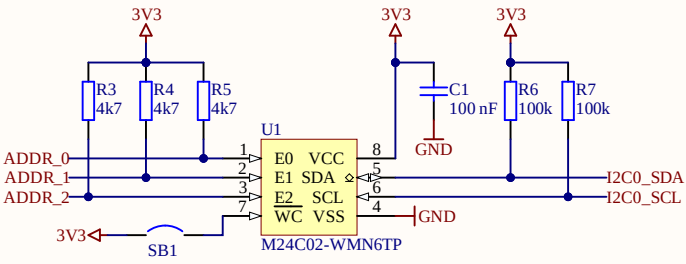
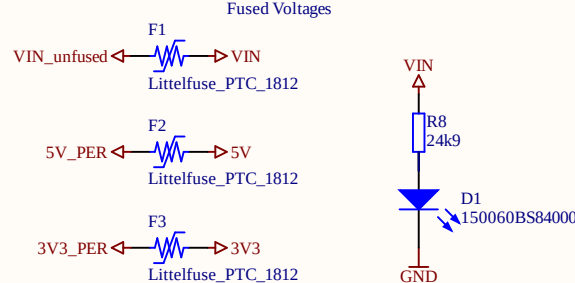
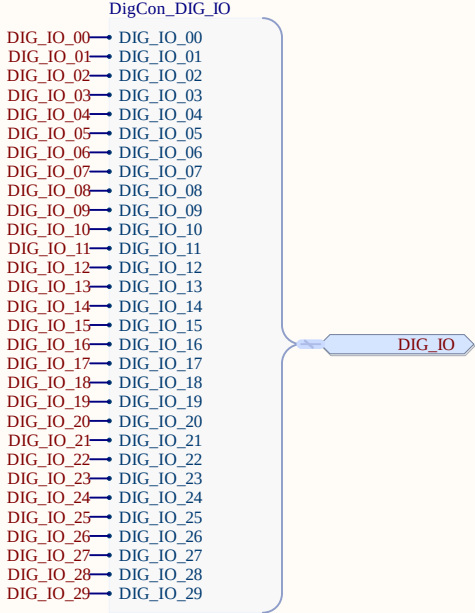
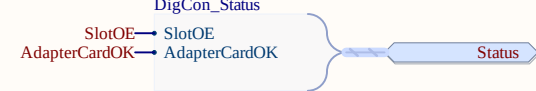
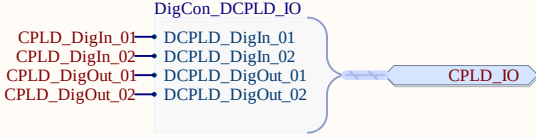
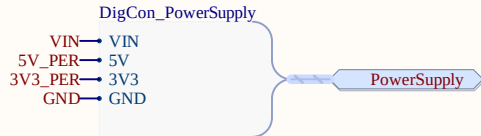
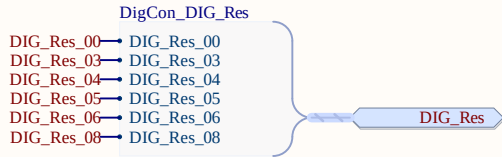
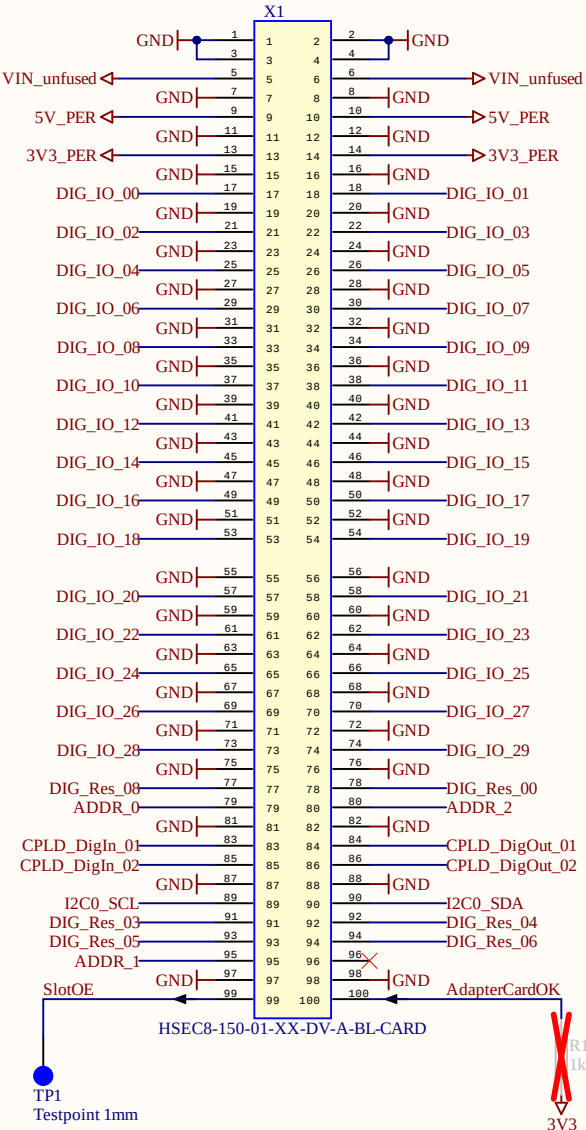


A

B

C

D



Note: place solder bump to protect EEPROM memory from write access.

PC-Address: 0b101 E2 E1 E0 = 0h5[0-7]

E2/E1/E0 is configured on UZ_CarrierBoard hardware related to the used slot D[1..5].

Title DigitalCarrierJack.SchDoc		UltraZohm www.ultrazohm.com Date: 03.04.2025 Sheet 2 of 5	
Revision: Rev02	Design Engineer: A. Geiger		
Project: UZ_D_Voltage_3V3_5V.PrjPCB			

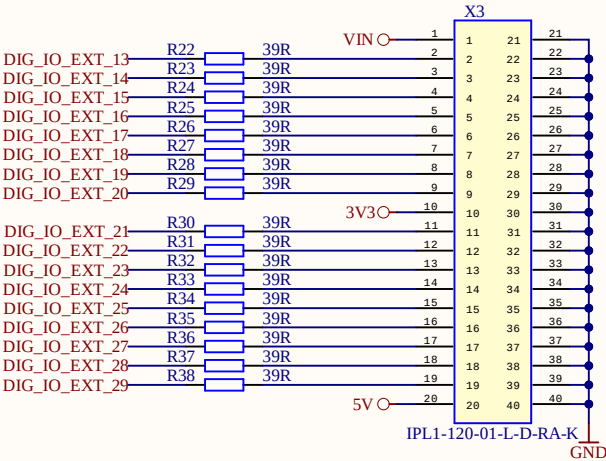
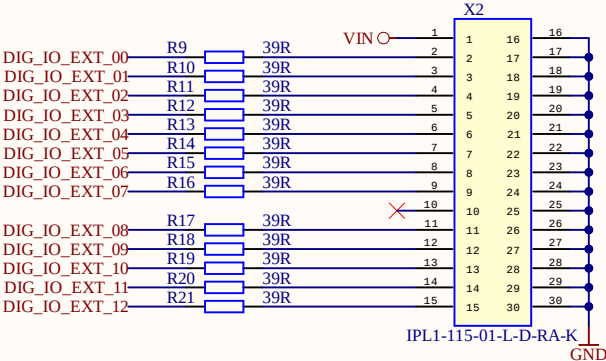
Pull-Down Resistors

Front Panel Connector

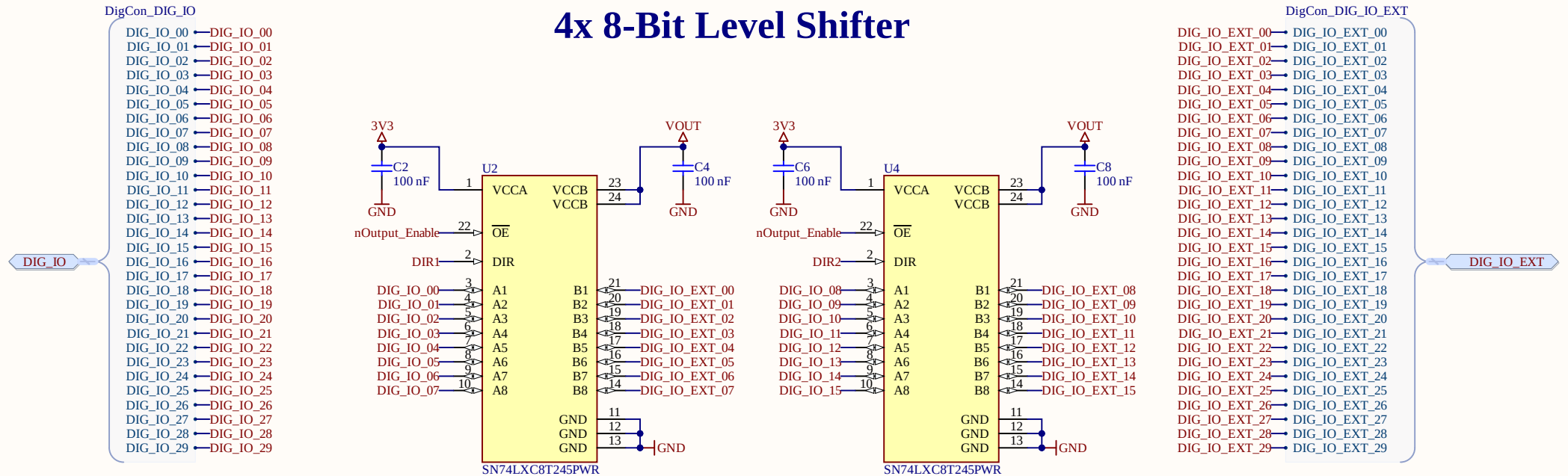
DigCon_DIG_IO_EXT

DIG_IO_EXT_00	DIG_IO_EXT_00
DIG_IO_EXT_01	DIG_IO_EXT_01
DIG_IO_EXT_02	DIG_IO_EXT_02
DIG_IO_EXT_03	DIG_IO_EXT_03
DIG_IO_EXT_04	DIG_IO_EXT_04
DIG_IO_EXT_05	DIG_IO_EXT_05
DIG_IO_EXT_06	DIG_IO_EXT_06
DIG_IO_EXT_07	DIG_IO_EXT_07
DIG_IO_EXT_08	DIG_IO_EXT_08
DIG_IO_EXT_09	DIG_IO_EXT_09
DIG_IO_EXT_10	DIG_IO_EXT_10
DIG_IO_EXT_11	DIG_IO_EXT_11
DIG_IO_EXT_12	DIG_IO_EXT_12
DIG_IO_EXT_13	DIG_IO_EXT_13
DIG_IO_EXT_14	DIG_IO_EXT_14
DIG_IO_EXT_15	DIG_IO_EXT_15
DIG_IO_EXT_16	DIG_IO_EXT_16
DIG_IO_EXT_17	DIG_IO_EXT_17
DIG_IO_EXT_18	DIG_IO_EXT_18
DIG_IO_EXT_19	DIG_IO_EXT_19
DIG_IO_EXT_20	DIG_IO_EXT_20
DIG_IO_EXT_21	DIG_IO_EXT_21
DIG_IO_EXT_22	DIG_IO_EXT_22
DIG_IO_EXT_23	DIG_IO_EXT_23
DIG_IO_EXT_24	DIG_IO_EXT_24
DIG_IO_EXT_25	DIG_IO_EXT_25
DIG_IO_EXT_26	DIG_IO_EXT_26
DIG_IO_EXT_27	DIG_IO_EXT_27
DIG_IO_EXT_28	DIG_IO_EXT_28
DIG_IO_EXT_29	DIG_IO_EXT_29

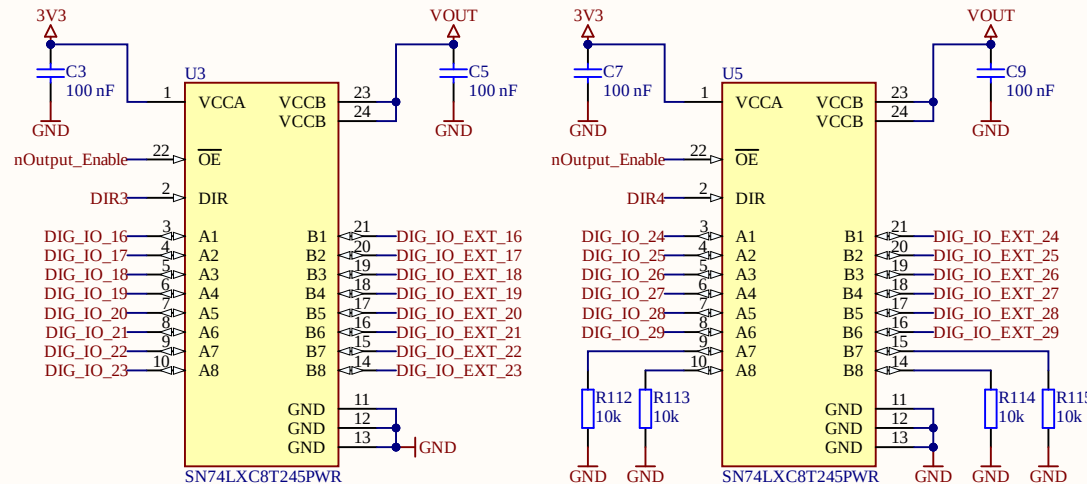
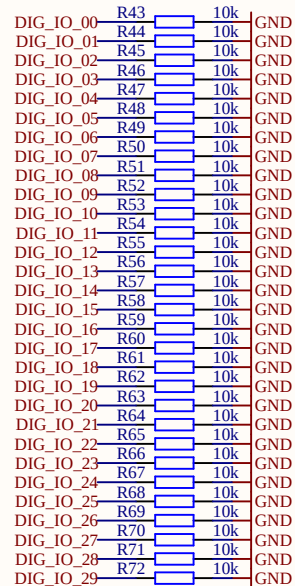
DIG_IO_EXT_00	R42	10k	GND
DIG_IO_EXT_01	R73	10k	GND
DIG_IO_EXT_02	R74	10k	GND
DIG_IO_EXT_03	R85	10k	GND
DIG_IO_EXT_04	R86	10k	GND
DIG_IO_EXT_05	R87	10k	GND
DIG_IO_EXT_06	R88	10k	GND
DIG_IO_EXT_07	R89	10k	GND
DIG_IO_EXT_08	R90	10k	GND
DIG_IO_EXT_09	R91	10k	GND
DIG_IO_EXT_10	R92	10k	GND
DIG_IO_EXT_11	R93	10k	GND
DIG_IO_EXT_12	R94	10k	GND
DIG_IO_EXT_13	R95	10k	GND
DIG_IO_EXT_14	R96	10k	GND
DIG_IO_EXT_15	R97	10k	GND
DIG_IO_EXT_16	R98	10k	GND
DIG_IO_EXT_17	R99	10k	GND
DIG_IO_EXT_18	R100	10k	GND
DIG_IO_EXT_19	R101	10k	GND
DIG_IO_EXT_20	R102	10k	GND
DIG_IO_EXT_21	R103	10k	GND
DIG_IO_EXT_22	R104	10k	GND
DIG_IO_EXT_23	R105	10k	GND
DIG_IO_EXT_24	R106	10k	GND
DIG_IO_EXT_25	R107	10k	GND
DIG_IO_EXT_26	R108	10k	GND
DIG_IO_EXT_27	R109	10k	GND
DIG_IO_EXT_28	R110	10k	GND
DIG_IO_EXT_29	R111	10k	GND



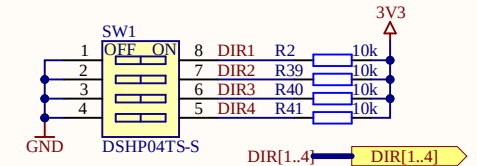
4x 8-Bit Level Shifter



Pull-Down Resistors



Direction per Level Shifter

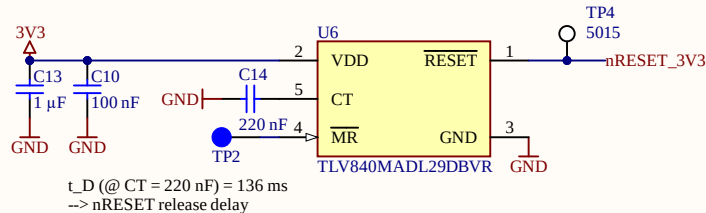
Table 8-1. Function Table⁽¹⁾

CONTROL INPUTS		Port Status		OPERATION
OE	DIR	A PORT	B PORT	
L	L	Output (Enabled)	Input (Hi-Z)	B data to A bus
L	H	Input (Hi-Z)	Output (Enabled)	A data to B bus
H	X	Input (Hi-Z)	Input (Hi-Z)	Isolation

(1) Input circuits of the data I/Os are always active and should be kept at a valid logic level.

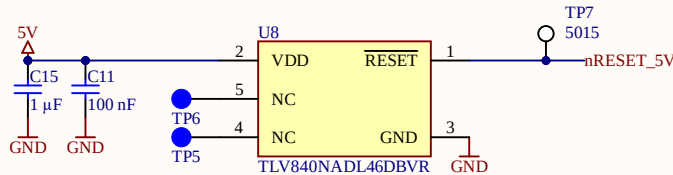
DIP-Switch	DIR	Signal Direction
OFF	HIGH	Output
ON	LOW	Input

3.3 V Voltage Monitoring



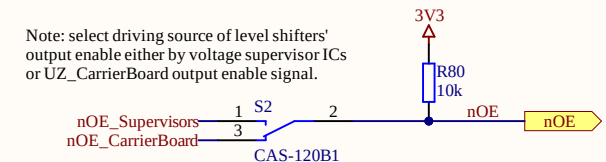
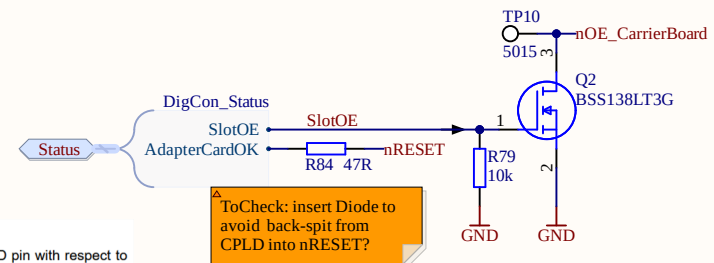
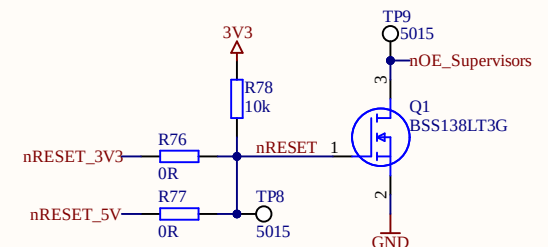
3V3 Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 2,9 V	LOW
2,9 ... 3,3 V	HIGH

5 V Voltage Monitoring



5V Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 4,6 V	LOW
4,6 ... 5 V	HIGH

Enable Signal Logic Inversion



Voltage Monitoring Functionality

8.4 Device Functional Modes

Table 8-1 summarizes the various functional modes of the device. Logic high is represented by "H" and logic low is represented by "L".

Table 8-1. Truth Table

VDD	MR	RESET	RESET
VDD < V _{POR}	Ignored	Undefined	Undefined
V _{POR} < V _{DD} < V _{IT}	Ignored	H	L
VDD ≥ V _{IT}	L	H	L
VDD ≥ V _{IT}	H	L	H
VDD ≥ V _{IT}	Floating	L	H

8.4.1 Normal Operation (V_{DD} > V_{POR})

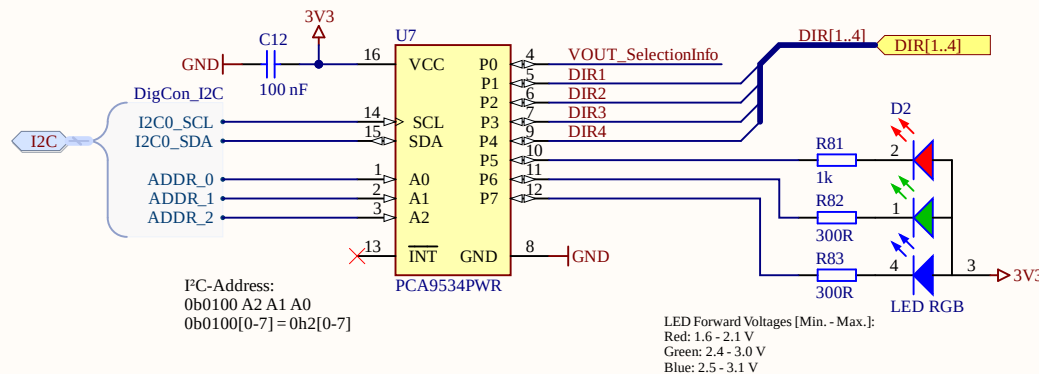
When VDD is greater than VPOR, the reset signal is determined by the voltage on the VDD pin with respect to the trip point (V_{IT}).

- MR high: the reset signal corresponds to VDD with respect to the threshold voltage.
- MR low: in this mode, the reset is asserted regardless of the threshold voltage.

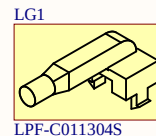
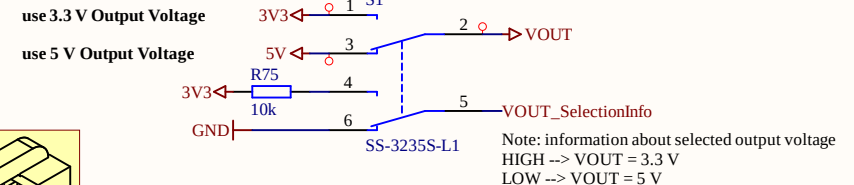
8.4.2 Below Power-On-Reset (V_{DD} < V_{POR})

When the voltage on VDD is lower than V_{POR}, the device does not have enough bias voltage to internally pull the asserted output low or high and reset voltage level is undefined.

I²C-Port Expander



Level Shifter - Output Voltage Selection



Title Output_Voltage_Selection_and_Enable.SchDoc		UltraZohm www.ultrazohm.com Date: 03.04.2025 Sheet 5 of 5
Revision: Rev02	Design Engineer: A. Geiger	
Project: UZ_D_Voltage_3V3_5V.PrjPCB		